[01] Lufeng Unit 1 (CGN)| Condenser modules lowered into position

Key words: CI MEH, condenser installation, integral module, jacking into position, modularisation

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(FCD+19.9 months) On **23 October 2026**, all **three integral modules** of the condenser in the conventional island of Lufeng Unit 1 were lowered into position, completing condenser installation. The complete assembly has a **maximum weight of approximately 742 tonnes**. The modules were **assembled off site, transported as integral units and jacked into position**, and their installation cleared the way for turbine generator installation and the hot functional test.



Three condenser modules in position in the conventional island at Lufeng Unit 1 (Image source: CGN).

The condenser modules are large and carry high self-weight loads, with stringent construction parameters and technical standards. The lower module, combined with its water chamber, weighs **508 tonnes**, and the upper module weighs **123.6 tonnes**. **Working space was limited between individual modules and between the modules and the building structures**, making lifting and transport complex and placing high demands on positioning accuracy.

The lower tube bundle, throat, water chamber and low-pressure heater were **integrated into a single assembly**, giving a **modularisation rate of 97%** and simplifying on-site work. Compared with the traditional method of **piece-by-piece lifting and on-site fit-up and welding**, which entailed **heavy site welding, risks from concurrent operations and difficult positioning control**, the approach **shortened the critical-path duration by about two months**.